

JSP3: Eval

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1 Introduction

This report evaluates the performance of the model `spec_qwen3:4b.json` using the gold standard dataset `spec_human.json`. The goal is to compare the model predictions with human annotations and measure how accurate the model is.

We focus on several aspects of evaluation. First, we calculate the overall accuracy of the model. Then, we analyze the results by different POS tag types and look at the most common errors. We also use WordNet to measure semantic similarity between gold and predicted synsets and compute near-miss scores. This helps us understand whether the model predictions are completely wrong or only slightly different in meaning.

2 Background

For semantic evaluation, we use WordNet through the NLTK library. WordNet is a lexical database of English words, where words are grouped into sets called synsets. Each synset represents one meaning of a word.

WordNet also stores relations between synsets, such as synonymy and hypernymy. These relations allow us to measure how close two meanings are. In this evaluation, WordNet is used to check whether the model predicts a synset that is semantically close to the correct one.

Resources

The WordNet corpus is provided by the NLTK library. It allows us to compute semantic similarity between synsets. In our evaluation, we use the Wu-

Palmer similarity measure, which compares synsets based on their position in the WordNet hierarchy.

3 Approach

The evaluation was located in the file `eval.py`. The input files are loaded using `argparse`. For each token, the model prediction is compared with the gold standard annotation.

We first compute the total accuracy. Then, we calculate accuracy for each POS tag type. Confusion matrices are created for POS tags and for synsets to analyze common mistakes. Precision, recall, and F1 scores are computed in both macro and micro variants.

To evaluate near-miss errors, we use WordNet from the NLTK library. Semantic similarity between synsets is computed using the Wu-Palmer similarity (`wup_similarity()`). We also calculate accuracy without the POS tag "x", because this tag causes many errors and strongly affects the final results.

4 Results

Accuracy

- Total accuracy: 57.87%
- Accuracy without "x" POS: 88.05% (5616/6378)

Breakdown by Tag

Table 1: Break down results by tag type

Tag	Correct	Count
a	75.60%	(914/1209)
dat:year	0.00%	(0/1)
loc	94.12%	(16/17)
n	64.18%	(1851/2884)
num	100.00%	(3/3)
org	50.00%	(1/2)
oth	50.00%	(1/2)
per	95.92%	(47/49)
r	75.26%	(569/756)
v	60.62%	(882/1455)
x	28.13%	(561/1994)

Confusion Matrix POS

Table 2: Confusion matrix POS

POS	MISSING	a	dat	dat:year	e	loc	n	num	org	oth	per	r	v	w	x	year	z
MISSING	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
a	31	1085	1	0	0	0	28	24	0	2	0	52	5	1	11	0	0
dat	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
dat:year	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
e	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
loc	0	0	0	0	0	16	0	0	1	0	0	0	0	0	0	0	0
n	51	77	17	0	0	31	2390	12	0	10	27	33	59	12	214	2	0
num	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0
org	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0
oth	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
per	0	0	0	0	0	0	2	0	0	0	47	0	0	0	0	0	0
r	7	15	8	0	0	0	13	1	0	0	0	679	2	1	37	0	0
v	50	2	0	0	0	1	37	0	0	0	0	4	1394	6	11	0	0
w	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
x	37	329	6	2	3	2	81	5	0	1	1	282	505	35	657	1	86
year	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
z	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Top Synset Confusions

Table 3: Top synset confusions

Gold	Model	Count
be.v.01	exist.v.01	21
miss.n.03	girl.n.01	17
be.v.05	be.v.01	17
state.v.01	say.v.08	13
hand.n.13	hand.n.01	11
exist.v.01	be.v.01	11
be.v.02	be.v.01	9
be.v.01	be.v.05	9
however.r.02	however.r.01	7
window.n.01	window.n.07	6

F-score Summary

Table 4: F1 Summary

Type	Precision	Recall	F1
Macro	0.482	0.703	0.513
Micro	0.734	0.73	0.734

Error Types

Table 5: Type of error

Type	Count
mismatch_pos	1977
generalized_tag	5
mismatch_offset	1127
coarse_tag	117
missing_target_tag	176
unknown_tag	301
missing_gold_tag	1

Near-miss Score

Table 6: Near-miss error examples

Gold	Model	Near-miss
summon.v.02	call.v.05	0.947
cat.n.01	big_cat.n.01	0.929
request.v.01	invite.v.07	0.909
costume.n.04	costume.n.01	0.900
edge.n.01	boundary_line.n.01	0.875

Near-miss Histogram

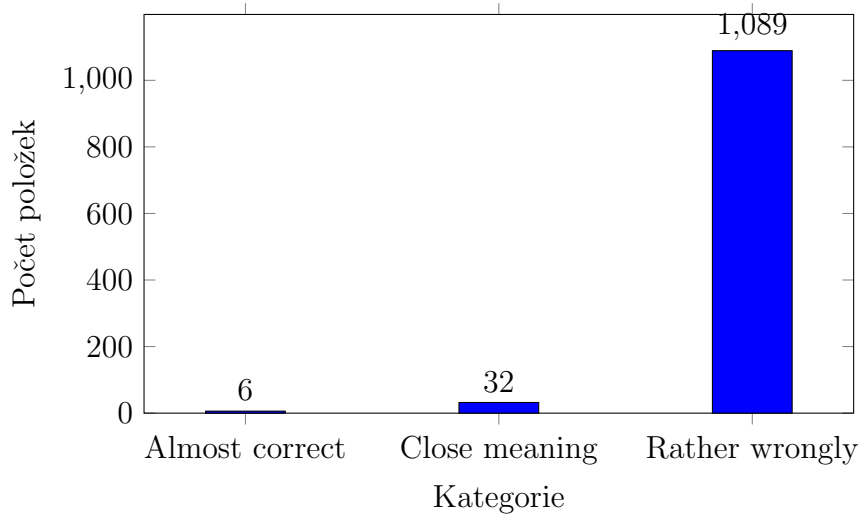


Figure 1: Histogram kvality Near-miss predikcí

Average near-miss score: 0.424

Near-miss weighted F1: 0.381

5 Discussion

Our evaluation shows that the model has overall accuracy of 57.87%. When we ignore the problematic POS tag "x", the accuracy increases to 88.05%. This means the model works well for normal and clear tokens, but unknown or unclear tokens make the performance worse.

Looking at tag results, the model is very good at numbers ("num"), person names ("per"), and locations ("loc"). It is worse at nouns ("n") and verbs ("v") and almost fails on very rare tags like 'dat:year'. This is probably because there are fewer examples of these tags in the dataset.

The confusion matrix and synset analysis show that the model often predicts similar but not exact meanings. Near-miss score (average 0.424) shows many mistakes are close in meaning. So the model understands the context, but not always the exact sense.

The main types of errors are POS mismatch and offset mismatch. This shows problems with tokenization or alignment with the gold data. Future work can focus on:

- Improving the handling of unknown POS tags ("x") and rare tags.

- Teaching the model to better distinguish similar meanings in synsets.

6 Conclusions

In this report, we evaluated the model `spec_qwen3:4b.json` on the dataset `spec_human.json`. The overall accuracy is 57.87%, and without the "x" tag it is 88.05%. The model works well on common and clear tokens, but rare and unclear tags are harder to predict.

The confusion matrix and near-miss scores show that many mistakes are semantically close. Future improvements can include better handling of rare tags, better distinction of similar meanings.